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Patient-Reported Outcomes

Adaptation of the Experimental Version of EQ-5D-Y-5L Into Bahasa Indonesia



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ABSTRACT

Objectives: The EQ-5D-Y-5L has been developed to provide an appropriate version of EQ-5D for children. This article reports the EQ-5D-Y-5L adaptation process into Bahasa Indonesia.

Methods: Forward and backward translations were performed to produce equal qualifiers between EQ-5D-Y-5L English and Bahasa versions. Unresolved issues were addressed in a cognitive debriefing, conducted with 10 children aged 8 to 15. A linguist proofread the final stage of the translation.

Results: Differences were found in wordings, headings, font styles, and qualifiers in EQ-5D English versions that require consideration when translating the English version into other languages. Two issues arose: first, levels 1, 2, and 4 in EQ-5D-Y-5L Bahasa matched the levels 1, 2, and 3 of EQ-5D-Y-3L, rather than levels 1, 3, and 4, as is the case in the English versions. Second, the translated words for “pain” and “worried,” used in the other EQ-5D Bahasa, were difficult to comprehend. Hence, we replaced these words with alternative Bahasa words at the cost of breaking consistency between the EQ-5D-Y-3L and EQ-5D-Y-5L Bahasa. There was an indication that the extensive cognitive debriefing caused response fatigue in children.

Conclusions: The adaptation process resulted in a comprehensible EQ-5D-Y-5L Bahasa Indonesia for children aged 8 to 16. Our findings suggest a need to review the existing EQ-5D-Y-3L and EQ-5D-3L in Bahasa Indonesia. While awaiting this review, research related to these 2 measures should be interpreted with caution in the context of international comparisons. Furthermore, we propose consideration of a more user-friendly approach to cognitive debriefing tailored for children.

Keywords: children, EQ-5D-Y-5L, HRQoL, instrument adaptation.

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Introduction

EQ-5D-Y is an instrument to measure health-related quality of life in children and adolescents. The instrument has been adapted from its parent version, the EQ-5D, by using age-appropriate wording and an improved layout to facilitate an instrument better suited for children aged 8 to 15.¹ The EQ-5D-Y descriptive system includes the same 5 dimensions as the adult version and uses 3 response levels per dimension (no problems, some problems, and a lot of problems). Similar to the EQ-5D-3L adult version, EQ-5D-Y has been criticized for its ceiling effects and insensitivity toward small health changes.^{2,3} Hence, an EQ-5D-Y version with an increased number of severity levels was developed, the EQ-5D-Y-5L. The experimental version of the instrument has been tested for its comprehensibility and feasibility. The benefit of having 5 levels compared with 3 in the EQ-5D youth instrument has been reported in several studies.⁴⁻⁶

Worldwide research has produced demands for the availability of EQ-5D-Y in other languages. Instrument translation is thus

required to produce linguistically and conceptually equivalent translations for different countries.⁷ Linguistic adaptation is especially important for EQ-5D instruments because each qualifier to reflect 5 different severity levels in the descriptive system will be independently weighted through the valuation process. Any inversion of qualifiers could lead to a threat to validity in producing a value set. In addition, for EQ-5D, as a global instrument used around the world, accurate and equal qualifiers across countries would allow comparison between different language groups.⁸ To serve this purpose, the EuroQol Group has developed a step-by-step guideline to translate EQ-5D questionnaires into other languages.⁹ A multistep approach to cultural adaptation, such as the one used by the EuroQol Group, can ensure that the translated versions are linguistically and conceptually equivalent.¹⁰ The guideline is updated on the basis of ongoing experience, which means that in this study, we have used the version which has now been revised.

This study is part of a larger investigation aimed at evaluating the validity of the EQ-5D-Y-5L in comparison with the EQ-5D-Y-3L

in Indonesia.^{5,11} A key initial step in this validation process is the translation of the EQ-5D-Y-5L into Bahasa Indonesia, ensuring that the instrument can be accurately understood by the children in Indonesia. Although EQ-5D-Y-3L has already been translated into Bahasa Indonesia by EuroQol office, the EQ-5D-Y-5L has not yet undergone this process. The aim of this study was to describe the adaptation process for translating the experimental English version of the EQ-5D-Y-5L into Bahasa Indonesia. By providing a comprehensive description of each step taken during the translation, this study aims to offer valuable insights and guidance for researchers involved in the translation of health-related quality-of-life instruments.

Methods

Instrument

EQ-5D-Y-5L

EQ-5D-Y is a generic instrument designed to measure health-related quality of life in young participants. The questionnaire has 5 dimensions: mobility (walking about), looking after myself, doing usual activities, having pain or discomfort, and feeling worried, sad, or unhappy. The wording and layout were adapted from the adult EQ-5D version to measure the cognitive levels and language of children aged 8 to 16 years old.¹

In its initial version, EQ-5D-Y had 3 severity levels. This version was extended to improve its sensitivity with respect to small health changes. Several phases were undertaken to develop a new descriptive system for EQ-5D-Y. In the first phase, a group of children aged 8 to 15 was interviewed to identify all possible child-friendly labels. Subsequently, the pooled labels were ranked to determine the relative severity of each label. The comprehensibility of the tested labels and the preferability toward 4-level severity or 5-level severity were identified during this phase. In the final phase, the selected labels were deliberated between countries to consider the translatability of the selected labels between languages. The new descriptive EQ-5D-Y-5L system has proved to be comparable between different language versions. The selected qualifiers for EQ-5D-Y-5L are as follows: no problems, a little bit of problems, some problems, a lot of problems, and cannot/extreme problems.

At the time this study was conducted, there was no official version of EQ-5D-Y-5L available from the EuroQol Group. Hence, an experimental version of EQ-5D-Y-5L was used. For forward translation purposes, the EQ-5D-3L Bahasa Indonesia, EQ-5D-5L Bahasa Indonesia, and EQ-5D-Y-3L Bahasa Indonesia versions were used in the comparisons.

Procedure

The instrument adaptation process was undertaken from November 2018 to February 2019. The first author led the translation process as project manager, following the EuroQol Group guidelines for “EQ-5D-Y Full Translation” version April 2018. A report was made on each translation step and reviewed by the Version Management Committee (VMC) of the Group. The translation process is shown in Figure 1.

Preparation: comparison between source versions

Before the translation process, comparing English EQ-5D source versions is required to identify the wording differences between versions. These comparisons are crucial in developing a comparable translation between an EQ-5D version in one language and between languages in one EQ-5D version. [Appendix 1](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.vhri.2025.101121> shows the delta (wording differences) between versions (ie, adult 3L vs adult 5L; EQ-5D-Y-3L vs EQ-5D-Y-5L; EQ-5D-Y-3L vs adult 3L; EQ-5D-Y-5L vs adult 5L).

[vhri.2025.101121](#) shows the delta (wording differences) between versions (ie, adult 3L vs adult 5L; EQ-5D-Y-3L vs EQ-5D-Y-5L; EQ-5D-Y-3L vs adult 3L; EQ-5D-Y-5L vs adult 5L).

Forward translation process

In the translation process, priority was given to achieving conceptual equivalence between the English and Bahasa Indonesia versions of the EQ-5D, as well as ensuring that the wording was comprehensible for children.

Forward translation. Forward translation from English to Bahasa Indonesia was performed by 2 qualified translators who were native speakers in Bahasa Indonesia and fluent in English. One translator was a PhD candidate in public health with a medical background, and the other had an economics background who was unfamiliar with health science. The translations were carried out independently.

Consensus meeting. A consensus meeting to discuss the translations was held between the translators and the project manager. A report was produced and discussed with the VMC.

Review by the VMC. The VMC compared the EQ-5D-Y-5L translation draft with the available version of EQ-5D in Bahasa Indonesia. Any discrepancies and challenges encountered during the translation process were discussed. Issues that remained unresolved were highlighted, and it was decided to test these directly with children during the cognitive debriefing phase. This phase resulted in the first consensus version of EQ-5D-Y-5L in Bahasa Indonesia.

Backward translation process

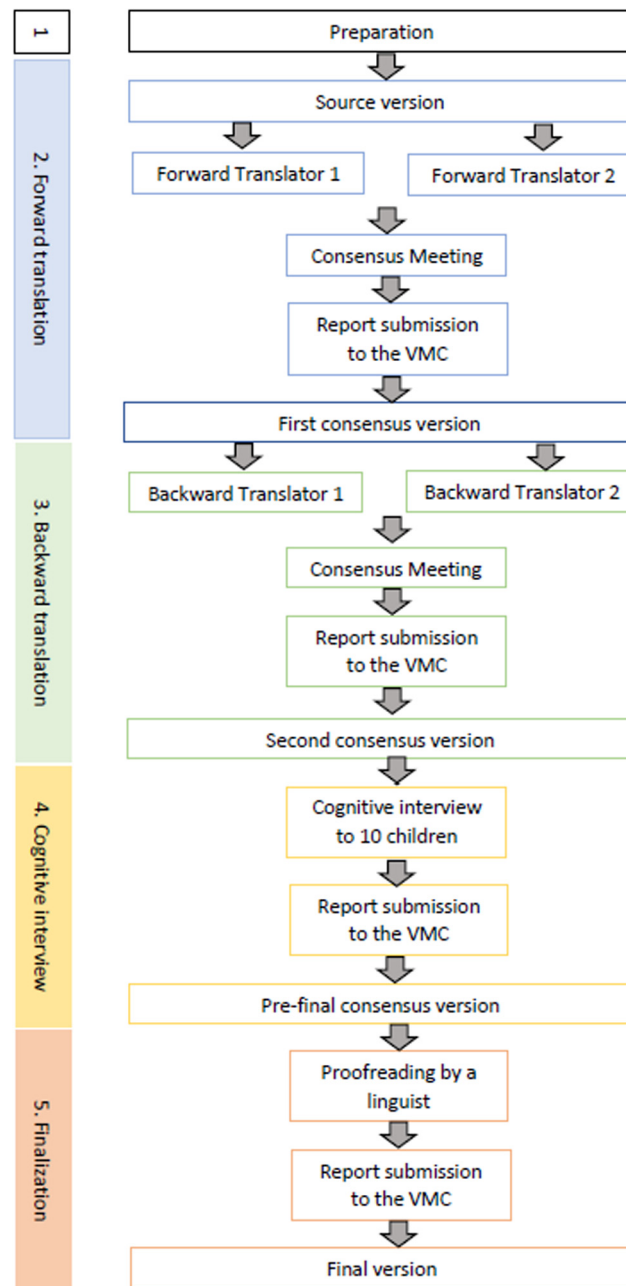
Backward translation. Translations from Bahasa Indonesia back into English were undertaken independently by 2 native English speakers who were also fluent in Bahasa Indonesia. One of the translators was a professional translator who had lived in Indonesia for years. The translator was tasked to translate the first consensus version back into English “in a fluent way.” A second translator studied Bahasa Indonesia for 6 years and had a language diploma, majoring in Indonesian studies. The second translator translated the instrument back to English in a literal way.

Consensus meeting and review by VMC. The fluent and literal back translations were discussed to find the best translation for the EQ-5D-Y-5L in Bahasa Indonesia. The second consensus version was produced in this stage.

Cognitive debriefing

The second consensus version was tested with 10 participants aged 8 to 15 years. The children were representative across age ranges (8-9, 10-12, and 13-15), with a particular emphasis on including younger children. Participants represented various economic statuses and health conditions and had never seen the questionnaire before. Healthy children were approached according to the interviewer convenience, whereas children with illnesses were approached at the hospital. All children were participating in school. The cognitive interviews aimed to evaluate the clarity, naturalness, and adequacy of wording and to confirm that the participants understood the wording intended.⁹ The cognitive interview comprised 3 sections: (1) the first inquired about the instructions and the overall impression of the questionnaire. (2) The second was a rating exercise, which required participants to rate the in-between qualifiers for levels 2, 3, and 4 (a little bit, some, and a lot

Figure 1. EQ-5D-Y-5L Indonesian version translation process based on the EuroQol translation guideline.



VMC indicates Version Management Committee.

of problems) of EQ-5D-Y-5L on a scale of 0 to 100. These scores were expected to reveal the hierarchical structure of the qualifiers used in the second consensus version. (3) The last section comprised an in-depth interview in which participants were prompted to describe their understanding of all the wordings used in the questionnaire. Additionally, participants were asked to provide examples of persons reflecting each severity level per dimension. A cognitive debriefing report was prepared and discussed with the VMC to achieve the prefinal version.

Finalization

The prefinal version was proofread by a linguist, a native speaker in Bahasa Indonesia who had not been involved in the

translation process. The main purpose of the proofreading was to ensure that there were no typographical errors (eg, grammatical and spelling mistakes) or layout errors in the final version of the questionnaire.⁹ A clean template and one with tracked changes were submitted for final approval with the proofreading report. The approved version was then referred as the “effective version.”

Results

Preparation: Comparing Source Versions of EQ-5D

Appendix 1 in Supplemental Materials found at <https://doi.org/10.1016/j.vhri.2025.101121> shows the differences between

the English source version of EQ-5D, the adult 3L, the adult 5L, the EQ-5D-Y-3L, and the EQ-5D-Y-5L. The new levels inserted in adult 5L were level 2 (slight problems) and level 4 (severe problems). For the youth versions, the new levels inserted were level 2 (a little bit of problems) and level 5 (cannot or extreme). All qualifiers representing the middle level of adult 5L use the word “moderate,” whereas the other versions use “some/moderate/a bit/quite.” Some adjustments were made for the headings, styles, and wordings used for the youth versions of EQ-5D (see [Appendix 1](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.vhri.2025.101121>).

[Appendix 2](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.vhri.2025.101121> shows the comparison of the Bahasa Indonesia version of EQ-5D 3L, the 5L, and EQ-5D-Y-3L. In the Bahasa Indonesia version of EQ-5D-Y-3L, level 3 uses the same qualifier as level 4 in the EQ-5D-5L Bahasa Indonesia, mirroring the English version. In the adult version of the 3L, there is no qualifier used to represent the problem’s magnitude at level 2. In detail: in level 2 of the 3L the translated sentence is “saya mempunyai kesulitan dalam...” Translated back into English, this sentence corresponds to “I have problems in...”

Forward Translation

Several conceptual and linguistic issues were identified from the forward translation process (see [Appendix 3](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.vhri.2025.101121>). Most of these issues were resolved during the first consensus meeting by comparing the translation with the existing version of EQ-5D in Bahasa Indonesia and the EuroQol guideline.¹² In discussion with the VMC, some of the conceptual and linguistic issues were identified and resolved. Unresolved issues were put forward for the cognitive debriefing sessions. All of this resulted in the first consensus version. The most prominent conceptual and linguistic changes made in this first consensus version are now discussed.

Linguistic issues

It was apparent from the outcomes of the forward translation process that the in-between levels qualifiers in EQ-5D-Y-5L do not fit comfortably in Bahasa Indonesia languages. Challenges arose in translating the word “some,” the qualifiers for level 3. “Some” can be translated as “sedikit,” “beberapa,” or “cukup.” “Sedikit” was not preferred because it also means “a little,” aligning with the qualifiers for level 2 in EQ-5D-Y-5L Bahasa Indonesia. The official Bahasa Indonesia dictionary defines “beberapa” in a countable manner: something more than 2 but not too many, which does not suit the context because the item pertains to the degree of problems rather than their frequency.¹³ “Cukup” was considered, although initially not preferred because of its definition as “having enough” in Bahasa, which did not express the intended meaning of the qualifiers.¹³ Other alternatives, such as “setengah,” and “kurang lebih” were peculiar translations very rarely used in a daily life. Consequently, “cukup” was chosen as the best alternative, aligning with level 3 in EQ-5D-5L.

Incorrect literal translations emerged during the translation process, which required adjustment during the consensus meeting. For example, “washing” was translated literally into “mencuci” (cleaning things with water¹³), whereas the intended definition of “washing” by the VMC is an ability to take a bath or shower.¹² The word “mandi” is used to represent “washing” in Bahasa Indonesia. A similar issue arises with “walking about” which can be literally translated as “berjalan sekitaran,” an uncommon phrase in Bahasa Indonesia. It was decided to omit the word “sekitaran” and retain the translation as “...kesulitan dalam berjalan.”

Conceptual issues

Pain/discomfort. The existing Bahasa Indonesia version of EQ-5D had translated pain as “kesakitan” (in EQ-5D-3L and EQ-5D-Y-3L) or “nyeri” (EQ-5D-5L). These 2 words do mean “pain”; however, the word “kesakitan” means experiencing extreme pain, whereas “nyeri” is not age appropriate for children. Hence, in the consensus meeting with the translators, a new word was proposed: “rasa sakit.” This acts as a noun and can be combined with different magnitudes of pain. The translated word for “discomfort” in Indonesia, can refer not only to physical pain but also to a psychological state, such as feeling unsuited in a specific environment. To avoid misconceptions, “pada tubuh” (body) was added as a descriptor for discomfort.

Worried/sad/unhappy. The translation of “worried” in EQ-5D-Y-3L Bahasa Indonesia mirrors the translation of “anxious” in the EQ-5D adult version, although in Bahasa Indonesia, “worried” and “anxious” are supposed to be translated differently. Anxious can be translated as “cemas.” This word is rarely used in a daily situation, especially in children. During a consensus meeting, a new word, “khawatir,” was proposed as the accurate translation of “worried.” To validate the proposed changes, the alternative translation of “worried” (“khawatir”) was tested in cognitive debriefing, compared with the existing translation (“cemas”).

Backward Translation

Issues were found during the translation of the qualifiers for level 2 and level 3. The level 2 qualifier in Bahasa Indonesia (“sedikit”) was back translated as “some” by the first translator and “a little” by the second translator, consistently over dimensions except for the worried/sad/unhappy dimension. Issues were also found for “cukup” as the qualifiers for level 3. None of the translators back translated “cukup” as “some.” The back translated phrase or word was “significant amount,” “fairly,” or “somewhat.” We decided to keep the qualifiers the same as in the first consensus version to align them with the other version of EQ-5D. The translation was tested for its clarity in the cognitive debriefing.

Cognitive Debriefing

Participants

The characteristics of the participants involved in the cognitive debriefing are presented in [Table 1](#). All participants agreed that, overall, the questionnaires and, in particular, the instructions were easy to understand. Improvements were made in the questionnaire’s layout and style because 3 participants commented on the visual presentation of the questionnaire. Notably, participants tended to focus on how to fill out the questionnaire (eg, using a tick ✓ or cross X), rather than on the time frame specified (... your health TODAY).

On the next task, the rating exercise, participants were instructed to rate levels 2, 3, and 4 on a scale from 0 to 100. Eight participants successfully arranged the level in the intended order from best to worst health in all dimensions (see [Appendix 4](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.vhri.2025.101121>). However, there were 3 to 5 children who gave almost similar scores for level 2 and 3 across dimensions, with a difference of less than or equal to 10 points. Notably, 1 child reversed the order of levels 2 and 3 in the mobility and self-care dimensions. These findings suggest that for some, children the distinction between the qualifiers for levels 2 and 3 were not that apparent. This finding was further supported in the next part of

Table 1. Characteristics of study participants.

Subject	Age (years)	Gender	Health condition	Urban/rural	Social economic status	Time to complete (minutes)	Health profile	VAS
R1	10	F	Healthy	Urban	Middle	2	11221	95
R2	11	M	Chronic illness	Urban	Middle to low	3	11111	95
R3	13	F	Chronic illness	Urban	Low	4	11111	75
R4	13	M	Chronic illness	Urban	Low	4	11111	95
R5	15	F	Chronic illness	Urban	High	3	11111	85
R6	9	F	Healthy	Urban	High	3	11111	100
R7	10	M	Healthy	Urban	Low	3	11111	50
R8	15	M	Healthy	Urban	Middle to low	3	11111	95
R9	9	F	Acute illness	Urban	Middle	3	11222	80
R10	8	M	Healthy	Urban	Middle	4	13231	90
Mean	11.3					3.2		

F indicates female; M, male; VAS, visual analog scale.

the cognitive debriefing task in which the children were asked to give examples for each level across all EQ-5D-Y-5L dimensions. Some children encountered difficulty in articulating distinct examples of persons with health problems at levels 2 and 3.

Across dimensions, participants found it easier to give examples of persons with mobility problems. For instance, 1 of the participants expressed “a little bit of problems” as “someone who has fatigue on their leg,” whereas “some problems” was expressed as “someone with a twisted ankle.” The severity differences between levels were apparent, and the responses given were varied. In contrast, in the “looking after myself” dimension, some children attributed the problems to things unrelated to health, such as “children who are not well-trained by their parents to take a bath, and need assistance.” Similar features emerged for examples given in the “doing usual activities” dimension. In the dimension “having pain or discomfort,” examples were given, such as someone with stomach-ache or having influenza, which indicated good comprehension of this dimension and its levels. It was also clear that the participants interviewed understood the “feeling worried or sad” concept, although the differentiation between all levels was sometimes not clear for them. The extensive methods of the cognitive debriefing appeared to evoke saturation in the responses reported by some participants.

As mentioned in the forward translation stage, there were 2 Indonesian words tested to represent the word “worried.” Seven out of 10 participants preferred the proposed translation of worried, which is “khawatir,” rather than the previous translation “cemas.” Participants found this word easier and familiar to use in everyday life. Participants found the instructions for the visual analog scale (VAS) scores easy to understand, and there were no issues related to the wording.

Results of the cognitive debriefing were discussed further with the VMC in producing the third consensus version of EQ-5D-Y-5L Bahasa Indonesia.

Finalization

The third consensus version was checked by a linguist, native in the Indonesian language, who confirmed that the questionnaire had no issues in terms of wordings and layout. This version was thus adopted as the final EQ-5D-Y-5L Bahasa Indonesia version.

Discussion

This study adapted the EQ-5D-Y-5L (experimental version) from English into Bahasa Indonesia. Using a meticulous multistep approach, the process commenced with forward translation, followed by backward translation, cognitive debriefing, and proof-reading. Extensive discussion with the VMC was conducted to ensure a high-quality translation of the EQ-5D-Y-5L Bahasa Indonesia version. Ultimately, a comprehensible Bahasa Indonesia version of EQ-5D-Y-5L (experimental) has been made available for children aged 8 to 16.

A prominent issue arose for the Bahasa Indonesia version of EQ-5D-3L and EQ-5D-Y-3L, which preceded the current translation of the youth 5-level version. It became apparent that having parallel wordings between versions of EQ-5D is an important component of the translation process. Similar qualifiers between 3L and 5L and between the adult and youth versions are encouraged to facilitate comparisons between data and samples. In contrast, EQ-5D-Y-3L Bahasa Indonesia has a different qualifier for level 2 compared with the other versions of EQ-5D Bahasa Indonesia. This may have happened because of the absence of a qualifier in EQ-5D-3L Bahasa Indonesia (“some,” “moderate,” or “a lot”) that acts as a reference for the EQ-5D-Y-3L translation. To establish parallel wording between the versions of Bahasa Indonesia EQ-5D, we thus suggest amendments for the qualifiers in EQ-5D-3L and EQ-5D-Y-3L Bahasa Indonesia. Such an improvement would be timely, given that there is not yet an available national Indonesian 3L value set: although an experimental value set for the Indonesian youth version has been established.¹⁴ As soon as a national value set is developed, the translation should be compatible with that research effort and the translation thus fixed in time.

Forward and backward translation is a prominent step in instrument adaptation aimed at achieving conceptual and semantic equivalence across cultures.^{15,16} In the forward translation process, the translators often leant toward a literal approach. To enhance suitability for children, certain adjustments are necessary to ensure that the wordings are familiar for children. It might be beneficial for the forward translation draft to include not only literal translations but also “simple” and fluent versions that consider children’s comprehension levels. This way, the literal and fluent translations can be compared during the consensus meeting to produce age-

appropriate wordings for the EQ-5D youth instruments. It is recommended that the EuroQol Group provide further guidance on forward translation in its translation guidelines.

Overall, although we are satisfied with the current Bahasa Indonesia version of the EQ-5D-Y-5L, caution is advised when administering this questionnaire. Some participants showed close ratings (less or equal to 10 points) between level 2 and level 3 across dimensions. One child with low economic status even showed level inversion in the mobility and self-care dimensions. This difficulty in discriminating between level 2 and level 3 is not language specific to Indonesia; similar challenges have been reported in other languages, such as in Malawi¹⁷ and Chinese.¹⁸ To enhance future translation studies of the EQ-5D-Y-5L, it would be beneficial to involve linguistic experts to refine the representation of the in-between levels qualifiers. Additionally, in terms of EQ-5D-Y valuation, particularly when involving adolescence,¹⁹ the rank order of qualifiers should be clearly explained during the valuation practice session to ensure accurate comprehension of their order.

Cognitive debriefing aimed to assess the level of comprehensibility and cognitive equivalence of the translation to the targeted population. During the cognitive debriefing sessions, children filled out the questionnaire before starting in in-depth interview. Apparently, there was a low correlation between the reported health profiles and the VAS scores reported by the children. Because we did not explore the rationale behind the reported VAS scores in detail with the children, an explanation for this discrepancy can only be speculative. The VAS asks children for a general assessment of their overall health, which may lead them to consider aspects of their health beyond those covered by the EQ-5D dimensions. All participants reported that they found the questionnaire easy to understand, suggesting that the health profiles and VAS scores were reported intentionally based on their comprehension of the questionnaire. This indicates that the children were able to understand the instrument effectively, even if the results seem inconsistent.

The rating exercise was the most helpful section of the cognitive debriefing in assessing comprehension for the 5 severity levels. Most children were able to give the desired responses indicated by the hierarchical ranking for the 5 severity levels in EQ-5D-Y-5L. In the in-depth interview section, some children struggled to give examples of people with specific health severities, especially at the end of the task. We found this task cognitively challenging for children, and its retention in the translation guidelines for the youth version of EQ-5D should be up for consideration. We believe that the new approach, the ranking exercise introduced by the VMC for EQ-5D youth translations, could be an answer to replace this task.^{17,20}

The representativeness of the study sample is one of the strengths of our study. Participants (1) came from low- and high-economic status families, (2) included younger and older children, and (3) involved healthy and chronically or acutely ill children. The final version of EQ-5D-Y-5L Bahasa Indonesia was also tested subsequently using a ranking exercise, the additional translation method for EQ-5D youth, and has been confirmed as comprehensible to children in Indonesia.²⁰

Conclusions

A hierarchical classification system of health that children can comprehend has been established for the EQ-5D-Y-5L Bahasa Indonesia version. The step-by-step translation process ensured the conceptual and linguistic equivalency of the Bahasa Indonesia EQ-5D-Y-5L with the English EQ-5D-Y-5L. Our study suggests the amendment of EQ-5D-Y-3L and EQ-5D-3L Bahasa

Indonesia. Furthermore, we propose the consideration of a more user-friendly approach to cognitive debriefing tailored for children.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.vhri.2025.101121>.

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